Row Space, Column Space and Null space

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Given an $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Consider the following m vectors of $\mathbb{R}^n$

$$r_1 = \begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{bmatrix}, r_2 = \begin{bmatrix} a_{21} \\ a_{22} \\ \vdots \\ a_{2n} \end{bmatrix}, \ldots, r_m = \begin{bmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mn} \end{bmatrix}$$

and the following n vectors of $\mathbb{R}^m$

$$c_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, c_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \ldots, r_n = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Then $r_1, r_2, \ldots r_m$ are called the row vectors of A and $c_1, c_2, \ldots, c_n$ are called the column vectors of A.

Row Space and Column Space

Definition(Row Space and Column Space)

Given an $m \times n$ matrix $A = [a_{ij}]$.

- The row space of A is the subspace of $\mathbb{R}^n$ spanned by the row vectors of A and is denoted by $\mathcal{R}(A)$. That is

$$\mathcal{R}(A) = \text{span}\{r_1, r_2, \ldots, r_m\}$$

- The column space of A is the subspace of $\mathbb{R}^m$ spanned by the column vectors of A and is denoted by $\mathcal{R}(A)$. That is

$$\mathcal{C}(A) = \text{span}\{c_1, c_2, \ldots, c_n\}$$

Example

Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$, then

$$\mathcal{R}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} \right\} = \left\{ \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + \alpha_2 \begin{bmatrix} 5 \\ 6 \\ 7 \\ 8 \end{bmatrix} \;:\; \alpha_1, \alpha_2 \in \mathbb{R} \right\}$$

and

$$\mathcal{C}(A) = \text{span}\left\{ \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \end{bmatrix}, \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \end{bmatrix} \right\}$$

$$= \left\{ \alpha_1 \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 6 \end{bmatrix} + \alpha_3 \begin{bmatrix} 3 \\ 7 \end{bmatrix} + \alpha_4 \begin{bmatrix} 4 \\ 8 \end{bmatrix} \;:\; \alpha_1, \alpha_2, \alpha_3, \alpha_4 \in \mathbb{R} \right\}$$

Definition(Row Equivalent)

Two matrices are row equivalent if one can be obtained from other by a sequence of elementary row operations.

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Example

The matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 4 \\ 5 & 8 \end{bmatrix}$ are row equivalent because

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} R_1 \to 2R_1 \begin{bmatrix} 2 & 4 \\ 3 & 4 \end{bmatrix} R_2 \to R_2 + R_1 \begin{bmatrix} 2 & 4 \\ 5 & 8 \end{bmatrix} = B$$

Theorem

If a $m \times n$ matrix A is row equivalent to a $m \times n$ matrix B, then the row space of A is equal to the row space of B.

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Theorem

If a matrix A is row-equivalent to a matrix B in row-echelon form (or reduced row echelon form), then the non-zero row vectors of B form a basis for the row space of A.

Process to find a basis for the row space of a matrix

Let A be a square matrix of size $m \times n$. Then to find a basis of row space of A:

- Convert the matrix A into its row echelon form (or reduced row echelon form).

- A basis for row space of A is the set of non-zero rows in row echelon form (or reduced row echelon form).

Example

Determine a basis and the dimension of the row space of the matrix

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix}$$

Solution

To get a basis of row space of A, we must reduce A to its row echelon form

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix} R_1 \leftrightarrow R_2 \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 3 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ \\ R_4 \to R_4 - R_1 \end{matrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 2 & -1 \\ 0 & 1 & 3 \end{bmatrix}$$

$$\begin{matrix} R_3 \to R_3 + 2R_2 \\ \\ R_4 \to R_4 + R_2 \end{matrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 4 \end{bmatrix} R_4 \to R_4 - 4R_3 \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

which is in row echelon form and therefore a basis for the row space of A is

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

this contains three elements and thus $\dim(\mathcal{R}(A)) = 3$.

Example

Determine a basis and the dimension of the row space of the matrix

$$A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & -1 & 0 & 0 & 2 \\ 2 & 1 & 6 & 0 & 1 \end{bmatrix} \begin{matrix} R_4 \to R_4 - R_1 \\ \\ R_5 \to R_5 - 2R_1 \end{matrix} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & -2 & -4 & -1 & 0 \\ 0 & -1 & -2 & -2 & -3 \end{bmatrix} \begin{matrix} R_4 \to R_4 + 2R_2 \\ \\ R_5 \to R_5 + R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & -1 & -2 \end{bmatrix} \begin{matrix} R_4 \to R_4 - R_3 \\ \\ R_5 \to R_5 + R_3 \end{matrix} \begin{bmatrix} 1 & 1 & 4 & 1 & 2 \\ 0 & 1 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which is in row echelon form and therefore a basis for the row space of A is

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 4 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

this contains three elements and thus $\dim(\mathcal{R}(A)) = 3$.

Process to find a basis for the column space of a matrix

There are two ways to find a basis for the column space:

(1). Find the row-echelon form of A. The columns from the original matrix which have leading entries when reduced to row echelon form gives a basis for the column space of A.

Process to find a basis for the column space of a matrix

There are two ways to find a basis for the column space:

(1). Find the row-echelon form of A. The columns from the original matrix which have leading entries when reduced to row echelon form gives a basis for the column space of A.

(2). The second way to find a basis for the column space of A is to recognize that the column space of A is equal to the row space of A^T. Finding a basis for the row space of A^T is the same as finding a basis for the column space of A.

Example

Find a basis and dimension of the column space of the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -1 \\ 3 & 6 & -1 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ \\ R_3 \to R_3 - 3R_1 \end{matrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix} R_3 \to R_3 - 2R_2 \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Leading entries corresponds to 1st and 3rd columns and therefore a basis for the column space of A is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \right\}$$

this contains three elements and hence $\dim(\mathcal{C}(A)) = 2$.

Solution (Another Solution)

Since column space of A is same as row space of A^t. Thus we'll find a basis for the row space of A^t and that will be a basis for the column space of A^t.

$$A^t = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ -1 & -1 & -1 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ \\ R_3 \to R_3 + R_1 \end{matrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix} R_2 \leftrightarrow R_3 \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus a basis for the column space of A is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

this contains three elements and hence $\dim(\mathcal{C}(A)) = 2$.

Rank of a Matrix

We have seen earlier that rank of a matrix is defined to be number of non-zero rows in its row echelon form (or reduced row echelon form). Here we'll see another definition.

Definition (Rank of a Matrix)

The rank of a matrix A is defined as following

$$\dim(\mathcal{R}(A)) = \text{rank}(A) = \dim(\mathcal{C}(A))$$

Nullspace

Definition (Nullspace)

If A is an $m \times n$ matrix A, then the Nullspace of A is denoted by $\mathcal{N}(A)$ and is defined as the set of all solutions to the homogeneous system of linear equations $Ax = 0$. i.e.

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n \ : \ Ax = 0\}$$

- The nullspace of an $m \times n$ matrix is a subspace of $\mathbb{R}^n$.
- The dimension of nullspace(A) is referred to as the nullity of A and is denoted by nullity(A).

Example

Find the Nullspace of the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

and hence find a basis and dimension of the nullspace.

Solution

The nullspace(A) is the set of all solutions to the homogeneous system of linear equations $Ax = 0$. We'll find solution using gausian elimination method.

$$A^* = \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 1 & 1 & 0 & 1 & 0 \end{array}\right] R_2 \to R_2 - R_1 \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & -1 & 0 \end{array}\right]$$

which is in row echelon form. The system corresponding to above row echelon form is

$$x_1 + x_3 + 2x_4 = 0$$
$$x_2 - x_3 - x_4 = 0$$

Set $x_3 = s$ and $x_4 = t$, then $x_2 = s + t$ and $x_1 = -s - 2t$.

Solution cont...

We can express the solution as

$$\mathcal{N}(A) = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \;:\; x_1, x_2, x_3, x_4 \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -s-2t \\ s+t \\ s \\ t \end{bmatrix} \;:\; s, t \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} -s \\ s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} -2t \\ t \\ 0 \\ t \end{bmatrix} \;:\; s, t \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} t \;:\; s, t \in \mathbb{R} \right\}$$

Thus a basis for $\mathcal{N}(A)$ is

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

and this contains two elements and hence $\text{nullity}(A) = \dim(\mathcal{N}(A)) = 2$.

Example

Find a basis for the nullspace of A, where

$$A = \left[\begin{array}{ccccc|c} 1 & 1 & 4 & 1 & 2 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 1 & 1 & 0 & 0 & 2 & 0 \\ 2 & 1 & 6 & 0 & 1 & 0 \end{array}\right]$$

We need to solve the system $Ax = 0$:

$$A^* = \left[\begin{array}{ccccc|c} 1 & 1 & 4 & 1 & 2 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 1 & 1 & 0 & 0 & 2 & 0 \\ 2 & 1 & 6 & 0 & 1 & 0 \end{array}\right] \begin{array}{c} R_4 \to R_4 - R_1 \\ \\ R_5 \to R_5 - 2R_1 \end{array} \left[\begin{array}{ccccc|c} 1 & 1 & 4 & 1 & 2 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & -2 & -4 & -1 & 0 & 0 \\ 0 & -1 & -2 & -2 & -3 & 0 \end{array}\right]$$

$$\begin{array}{c} R_4 \to R_4 + 2R_2 \\ \\ R_5 \to R_5 + R_2 \end{array} \left[\begin{array}{ccccc|c} 1 & 1 & 4 & 1 & 2 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & -1 & -2 & 0 \end{array}\right] \begin{array}{c} R_4 \to R_4 - R_3 \\ \\ R_5 \to R_5 + R_3 \end{array} \left[\begin{array}{ccccc|c} 1 & 1 & 4 & 1 & 2 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

which is in row echelon form. The system corresponding to above row echelon form is

$$\begin{aligned} x_1 + x_2 + 4x_3 + x_4 + 2x_5 &= 0 \\ x_2 + 2x_3 + x_4 + x_5 &= 0 \\ x_4 + 2x_5 &= 0 \end{aligned}$$

Set $x_3 = s$ and $x_5 = t$ then $x_4 = -2t$, $x_2 = -2s + t$ and $x_1 = -2s - t$. The solution set to the system is given by

$$\mathcal{N}(A) = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} : x_1, x_2, x_3, x_4, x_5 \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -2s - t \\ -2s + t \\ s \\ -2t \\ t \end{bmatrix} : s, t \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} -2s \\ -2s \\ s \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -t \\ t \\ 0 \\ -2t \\ t \end{bmatrix} : s, t \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} s + \begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} t : s, t \in \mathbb{R} \right\}$$

Thus a basis for $\mathcal{N}(A)$ is

$$\left\{ \begin{bmatrix} -2 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

and this contains two elements and hence nullity$(A) = \dim(\mathcal{N}(A)) = 2$.

Rank-Nullity theorem

Rank-Nullity theorem

Let A be an $m \times n$ matrix. Then the rank-Nullity theorem states that

$$\text{rank}(A) + \text{nullity}(A) = n$$

Example

Verify rank-nullity theorem for following matrix

$$A = \begin{bmatrix} 3 & 6 & 6 & 3 & 9 \\ 6 & 12 & 13 & 0 & 3 \end{bmatrix}$$

Solution

The augmented matrix of the homogeneous system of linear equations $Ax = 0$:

$$A^* = \left[\begin{array}{ccccc|c} 3 & 6 & 6 & 3 & 9 & 0 \\ 6 & 12 & 13 & 0 & 3 & 0 \end{array}\right] R_1 \to \frac{R_1}{3} \left[\begin{array}{ccccc|c} 1 & 2 & 2 & 1 & 3 & 0 \\ 6 & 12 & 13 & 0 & 3 & 0 \end{array}\right]$$

$$R_2 \to R_2 - 6R_1 \left[\begin{array}{ccccc|c} 1 & 2 & 2 & 1 & 3 & 0 \\ 0 & 0 & 1 & -6 & -15 & 0 \end{array}\right] R_1 \to R_1 - 2R_2$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 13 & 33 & 0 \\ 0 & 0 & 1 & -6 & -15 & 0 \end{array}\right]$$

which is in reduced row echelon form and hence $\operatorname{rank}(A) = 2$. *The system corresponding to above rref is*

$$x_1 + 2x_2 + 13x_4 + 33x_5 = 0$$
$$x_3 - 6x_4 - 15x_5 = 0$$

Solution cont ...

Set $x_2 = r$, $x_4 = s$ and $x_5 = t$ then $x_1 = -2r - 13s - 33t$, $x_2 = r$, $x_3 = 6s + 15t$, $x_4 = s$ and $x_5 = t$. Therefore the nullspace of A is

$$\mathcal{N}(A) = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} : \; x_1, x_2, x_3, x_4, x_5 \in \mathbb{R} \right\} = \left\{ \begin{bmatrix} -2r - 13s - 33t \\ r \\ 6s + 15t \\ s \\ t \end{bmatrix} : \; r, s, t \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} -2r \\ r \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -13s \\ 0 \\ 6s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} -33t \\ 0 \\ 15t \\ 0 \\ t \end{bmatrix} : \; r, s, t \in \mathbb{R} \right\}$$

$$= \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \\ 1 \end{bmatrix} t : \; r, s, t \in \mathbb{R} \right\}$$

Solution cont . . .

Thus a basis of $\mathcal{N}(A)$ is

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -13 \\ 0 \\ 6 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -33 \\ 0 \\ 15 \\ 0 \\ 1 \end{bmatrix} \right\}$$

which contains three elements and hence

$$\text{nullity}(A) = \dim(\mathcal{N}(A)) = 3$$

Now,

$$\text{rank}(A) + \text{nullity}(A) = 2 + 3 = 5 = \text{No. of columns in } A$$

Hence, rank-nullity theorem is verified.

Exercise 1:

For each of the following matrices, find a basis for the row space, a basis for the column space, and a basis for the null space and hence verify rank-nullity theorem.

(a) $\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 4 & 7 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} -3 & 1 & 3 & 4 \\ 1 & 2 & -1 & -2 \\ -3 & 8 & 4 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 3 & -2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 4 & 5 & 6 \end{bmatrix}$

Exercise 2:

Find bases for $\mathcal{R}(A)$ and $\mathcal{N}(A)$ of the matrix

$$\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$$

Also find a basis for $\mathcal{C}(A)$ by finding a basis for $\mathcal{R}(A^T)$.

Exercise 3:
Find the nullity and the rank of each of the following matrices:

(a) $\begin{bmatrix} 1 & 3 & 1 & 7 \\ 2 & 3 & -1 & 9 \\ -1 & -2 & 0 & -5 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 2 & 1 & 5 & -2 \end{bmatrix}$

For each of the matrices, show that $\dim \mathcal{R}(A) = \dim \mathcal{C}(A)$ directly by finding their bases.

Exercise 4:

Let $A = \begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 3 & 0 & 5 \\ 1 & 1 & 6 & 3 \end{bmatrix}$.

(a) Find a basis for the nullspace of A.

(b) Find a basis for the row space of A.

(c) Find a basis for the range of A that consists of column vectors of A.

Exercise 5:

Let $A = \begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 3 & 0 & 5 \\ 1 & 1 & 6 & 3 \end{bmatrix}$.

(a) Find a basis for the nullspace of A.

(b) Find a basis for the row space of A.

(c) Find a basis for the range of A that consists of column vectors of A.